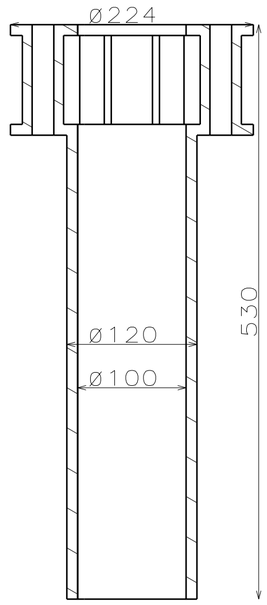


End view (1:4)



SectionA-A (1:4)

DESIGN DATA / TASARIM VERILERI (cad/params/carrier.txt)
Planets: 4
Planet radius [mm]: 82
Pin = inner race [mm]: 38
Pin length [mm]: 82
Plates: diameter x thickness [mm]: 224 x 10
Datums: A = outer bearing seat, B = inner bearing seat (axis A-B); C = carrier rear face
Dimensions on the views are the main sizes from the model; all other sizes are in the tables.

TOLERANCES AND CALLOUTS / TOLERANSLAR (tolerances_v1.csv; GD&T frames as this table, R19)
NU1024 journals (dia 120), datum axis A-B: cylindricity = 0.008 mm | A, B
front plate inner face, datum C: perpendicularity = 0.010 mm | A-B
four planet pin bores (dia 38): position (diametral zone) = 0.010 mm | A-B|C
planet pin bores: parallelism to A-B over the pin length = 0.005 mm | A-B
QJ1024 seat and shoulder: total runout (axial) = 0.015 mm | A-B
flange P115 pilot and face: total runout = 0.020 mm | A-B
pin in the plate (dia 38): ISO 286 fit = H6 / s6
catcher-lip axial gap at 20 degC: gap = 0.80-1.00 mm

MATERIAL AND PROCESS / MALZEME VE PROSES (materials_v1.csv, S43)
Material: AISI 4340, vacuum arc remelted
Specification: AMS 6414
Condition: Quenched and tempered to 1240 MPa tensile (AMS 2759/1); splines of the sun quill and the transfer sleeve bore gas nitrided (AMS 2759/6, TBR)
Hardness and case: 38-42 HRC core; nitrided splines at least 600 HV (TBR)
Surface texture: Bearing journals ground Ra <= 0.4 um; fillets polished; carrier pin bores line-bored to the 0.010 mm true position of step 2.4
NDT: Ultrasonic of the forging (AMS 2630), magnetic particle (ASTM E1444) after final machining
Protection: Oil-wetted surfaces bare; the propeller shaft outside the nose and the flange: zinc-nickel (AMS 2417) with hydrogen embrittlement bake, epoxy primer and topcoat (ENV-03)

CRITICAL PART / KRITIK PARCA (CS-E 515; safety/critical_parts_v1.csv)
Mark: CRITICAL PART - CS-E 515 / KRITIK PARCA; serial number on the part; life tracking: serial number on the part, flight cycles and hours counted by the engine monitoring unit (1 cycle per flight, plus each LD-01 exceedance), life limit in the ICA airworthiness limitations section
Life limit: 20,000 flight cycles declared; basis: the GAG stress range at each governing feature is below the -3 sigma crack-initiation endurance of that material and process (cited data for AISI 4340 VAR forgings, AMS 6414)
NDT at manufacture: UT of the forging (AMS 2630 class AA), MPI after final machining as the general surface check (NER 21); eddy current (ET, NASA-STD-5009C sizes) of the bearing seat, the thread root and the pin bore corners
NDT at TBO: ET of the bearing seat, the thread root and the pin bore corners at each TBO; the pin bores with the pins pressed out (new pins fitted after) or by a face-corner procedure with the pins in place; open NDE item: demonstration of the procedures, probability of detection at the thread root and the pin bore corners (3.10, 3.11); NDT at TBO after stripping the coating (NASA-STD-5009C sizes credited only under NER 12, 13 and 21, otherwise a demonstrated POD)
D4 governing locations: Propeller shaft P103 at the outer bearing seat; Propeller shaft P103 at the KM24 nut thread root; Carrier of P103 at a pin bore edge; Propeller shaft P103 at the shoulder-ring groove

NOTES / NOTLAR
1. Planet pins P104 are separate M50NiL parts, pressed in (LN2 shrink) and retained (3.4; materials_v1.csv M5N); pin bores machined before the pins are fitted.
2. R5 at the shaft-to-front-plate corner (3.9 brief); polish the radius, no tool marks across it.
3. QJ pull-out shoulder-ring groove: root diameter 118 mm, full radius, 1 mm deep (3.7 D4 location 12, OI-16).
4. Catcher-lip collar P118 shrink fit at z 512-516 on the same setup as the QJ1024 seat (gdt_v1.csv).
5. CAD views from the Phase 2 model; the tables and notes govern where they differ (3.9 CAD update open).

PLANET CARRIER WITH PROPELLER SHAFT P103 / TASIYICI VE PERVANE MILI P103
Part / Parca: RGB-P103
Material / Malzeme: AISI 4340 VAR (AMS 6414); pins P104 M50NiL (AMS 6278)
Scale / Olcek 1:4 mm ISO 2768-m unless noted; ASME Y14.5-2018
Turboprop RGB, step 3.9 representative drawing / adim 3.9 temsili resim
Hamit Tanboga values: manufacturing/drawing_checklist_v1.csv
Sheet / Pafta 1/1 GD&T as the callout table (tolerances_v1.csv)